Piecewise Vehicle Performance Modeling from Onboard Measurements

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Abstract—Three Jetta and three Camry test records supply piecewise models for four driving courses. Linear fits to filtered acceleration give quadratic speed and cubic position through integration. Averaging trial curve fit parameters produces one parameter set per car, including separate RPM and fuel-rate relationships. The models predict 16.12 s and 17.86 s for Jetta and Camry on the 100-yard course, and 21.77 s and 26.40 s on the 200-yard course with braking.

I. Introduction

The Purpose of this project was to collect real vehicle performance data and use that data to create mathematical models that will predict how vehicles behave under different driving conditions. We had to use an OBD-II reader to connect to a vehicle to record information while driving the car from a standing idle to 35 mph and braking from 35 mph back to a still idle. The recorded data included variables such as time per trial, speed (MPH), engine RPM (revolutions per minute), and fuel consumption (for all trials) etc. The data was then imported into MATLAB to create graphs and mathematical models depicting the data and predictions using the data.

We compared the data of our two chosen vehicles (Volkswagen Jetta and Toyota Camry) on small, controlled courses to hopefully be able to predict how the vehicles may perform on other required courses. Through the data we collected previously, we can predict travel time for the vehicle on possible paths, distance traveled, engine RPM, and fuel use for courses that were not measured directly [1].

The goal of the project was to use the measured OBD-II data to develop a model for different course's position, speed, acceleration, RPM, and fuel use. Then we will use these models to predict vehicle performance under different courses. Our predictions assume that driver input will stay the same during the other courses. Lastly, these models could also be extrapolated to predict the Irwindale drag strip(1/4mile acceleration course).

II. Background, Measurements, and Processing

The Vehicle motion in this project was analyzed by the relationship between acceleration, velocity, and position. Velocity describes how the position changes with respect to time, and acceleration describes how velocity changes with time. Due to differentiating on measuring speed data to find acceleration, a Kalman filter was used to decrease noise and create a smoother acceleration from OBD-II data. From Figure 1, we could see the relationship between acceleration, velocity, and position. Where Acceleration can be integrated to find out the velocity, and velocity could be integrated to find position.

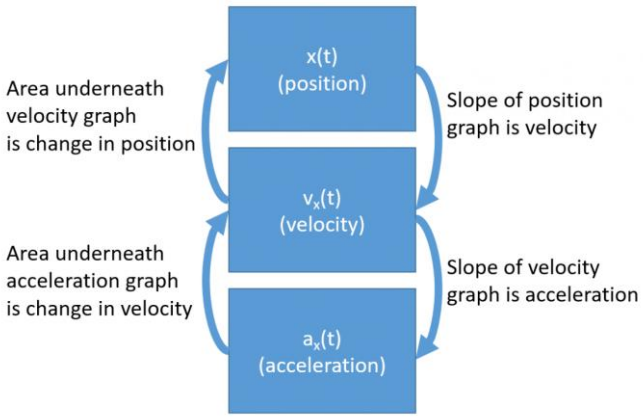


Figure 1. Relationship between Acceleration, Velocity, and Position. [6]

In Figure 2, the vehicle motion had been separated into different phases, including idle, acceleration, cruising, and braking. Each phase represents a different type of motion. During idle, the vehicle has zero velocity and zero acceleration. During acceleration, the vehicle velocity increases over time, and velocity decreases during breaking. Position, velocity, and acceleration are connected because change acceleration affects velocity, and change in velocity affects the vehicle position over time.

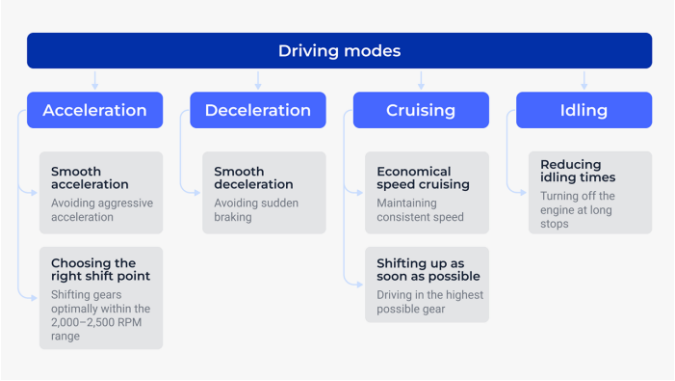


Figure 2. Different phase of vehicle motion. [7]

Engine RPM and fuel consumption are also related to vehicle motion. As vehicle speed changes, engineer RPM changes because engine and wheel are connected though the drivetrain. Speaking of power going from the engine to the wheels, we can remember that energy is lost as it goes through the drivetrain. Different acceleration could produce different RPM and fuel-use patterns.

We began first by measuring the data from the vehicles using an OBD II reader, and the online application OBD Fusion. We toggled only the data necessary for our needed calculation. We found having too many types of data being collected at the same

time can slow the speed of collection. So, we stuck to a happy middle in terms of necessary points of data collection and less necessary more well-rounded types of data to use to check our outputs. For each trial, we followed these steps: (1) connect the OBD-II reader and start recording in OBD Fusion with the vehicle stationary and the engine idling; (2) press the throttle gently and accelerate smoothly from 0 to 35 mph; (3) brake smoothly from 35 mph to 0 mph; and (4) stop recording and save the log. We aimed to use similar throttle and braking inputs between trials; pedal position and force were not measured. OBD Fusion records from the application were then imported into MATLAB. We collected the data from six similar trials to the procedure described above. Three of the trials we collected were from a 2022 Volkswagen Jetta SE, and the remaining three were collected from a 2024 Toyota Camry SE. Table I summarizes the six retained logs [2]. Important to note trials for the Volkswagen Jetta were recorded at a bit under 10hz and trials for the Toyota Camry were recorded at about 20hz.

TABLE I
RECORDED TRIAL COVERAGE

Trial	Rows	Span (s)	Rows/s	Peak (mph)
J1	156	18.84	8.23	34.80
J2	130	16.40	7.86	36.66
J3	141	16.56	8.46	37.28
C1	481	24.01	19.99	35.42
C2	510	25.45	20.00	34.18
C3	457	22.80	20.00	34.80

J = Jetta; C = Camry. N is the number of logged rows. Rows/s = (N − 1)/elapsed time, not the independent refresh rate of each signal.

Jetta records include time, speed, acceleration, engine RPM (revolutions per minute), trip distance, trip fuel, and MPG (miles per gallon). Camry records provide time, speed, engine RPM, and fuel mass rate. Speed is converted from mph to ft/s using 5280/3600. Camry distance is calculated with cumtrapz(time, speed); Jetta distance and fuel plots use changes in the logged trip counters. The original records remain unchanged.

The motion-section function uses speed to identify initial idle, acceleration, deceleration, and final idle when present. Each phase uses local time $\tau = t - t_start$: t is the current elapsed trial or course time and t_start is the phase start time, so τ is time elapsed within that phase (all in seconds). Recorded speed and position for all three trials of each car are provided in the appendix; dots represent samples, including calculated estimates.

Discrete differentiation amplifies noise. This is because when doing discrete differentiation, the smaller the time interval we have, the smaller the denominator will be, which can amplify a small noise into a large change in results. So we used a Kalman filter that estimates speed and acceleration together, predicting with a constant acceleration during a small time interval so that the filter smooths the graph and therefore corrects the measured speed [3] and predicted acceleration. We fit its smoother acceleration estimates rather than a direct numerical derivative, because overall the filtered estimates are more accurate and straightforward to the actual movement of our vehicles, while the direct numerical derivative is noisy and less predictable.

Figs. 3–4 show the filtered acceleration for the Camry and Jetta trials.

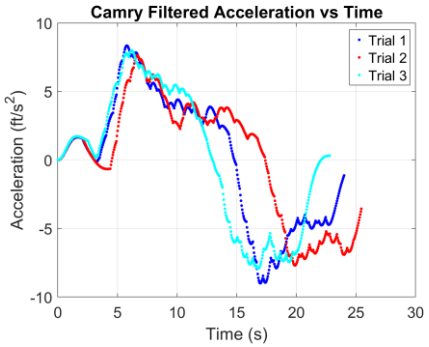


Figure 3. Camry Filtered Acceleration vs. Time. Filtered acceleration estimated from vehicle speed for the three Camry trials, showing acceleration and braking.

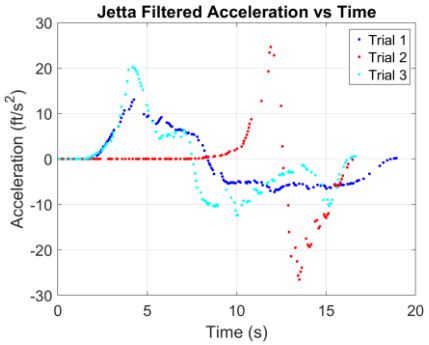


Figure 4. Jetta Filtered Acceleration vs. Time. Filtered acceleration for the three Jetta trials, showing acceleration and braking behavior.

III. Piecewise Models

For each acceleration or braking interval, polyfit(τ, filtered_accel, 1) returns [j, a_0]. Here a_0 is acceleration at the phase origin in ft/s² and j is jerk in ft/s³. We average the three coefficient vectors with equal weight, then integrate the mean acceleration using the course boundary values [4]:

$$a(\tau) = a_0 + j\tau \quad (1)$$

$$v(\tau) = v_0 + a_0\tau + (j/2)\tau^2 \quad (2)$$

$$x(\tau) = x_0 + v_0\tau + (a_0/2)\tau^2 + (j/6)\tau^3 \quad (3)$$

Here $a(\tau)$, $v(\tau)$, and $x(\tau)$ are acceleration (ft/s²), speed (ft/s), and position (ft). The subscript 0 denotes the phase-start value: a_0, v_0, and x_0. Jerk $j = da/d\tau$ is the rate of change of acceleration (ft/s³). Phase time resets, but position retains the common trial or course origin.

TABLE II
MEAN ACCELERATION PARAMETERS

Phase	a_0 (ft/s²)	j (ft/s³)
J accel	1.9925	1.6291
J brake	-3.0423	-1.8189
C accel	1.9667	0.2394
C brake	-3.8215	-0.2047

MATLAB polyint supplies the integrated coefficients, and polyval evaluates each phase [4]. At every transition, the preceding phase supplies the next initial position and speed. Cruising has a = 0, v = vc, and x = x0 + vcτ. At idle, speed and acceleration are zero and position is constant. The four course

clocks begin at launch. The 15 s idle intervals displayed on RPM plots are excluded from the reported course times and fuel totals.

Figs. 5–7 illustrate Eqs. (1)–(3) using Course 2. Within each acceleration or braking phase, acceleration is linear, speed is quadratic, and position is cubic. The cruising portion has constant speed.

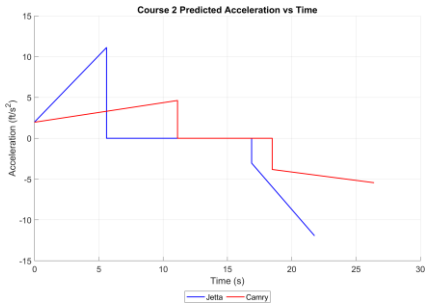


Figure 5. Course 2 Predicted Acceleration vs. Time. Predicted acceleration during the acceleration, cruising, and braking phases of Course 2.

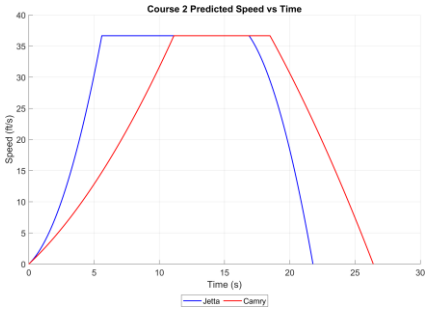


Figure 6. Course 2 Predicted Speed vs. Time. Predicted vehicle speed through acceleration, cruising, and braking in Course 2.

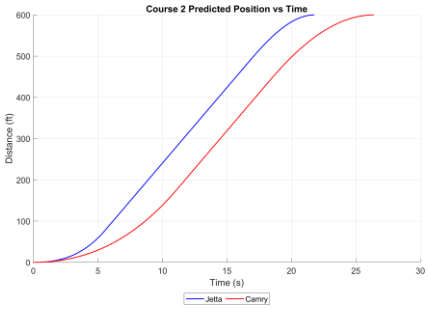


Figure 7. Course 2 Predicted Position vs. Time. Predicted distance traveled by the Jetta and Camry during Course 2.

Engine RPM and fuel rate are fitted against speed separately during acceleration and braking. Each trial uses a first-order polyfit; the three slope/intercept pairs are averaged. qf(t) is the instantaneous fuel consumption rate at time t (sec), F(t) is the cumulated fuel usage at time t; let v_mph = v × 3600/5280 be speed in mph, distinct from v in ft/s (max() function is used to clamp the values to reasonable range, for example, rpm must be larger than idle rpm unless engine is dead, and fuel rate must be positive):

$$\mathrm{RPM}(v_\mathrm{mph}) = \max(\mathrm{RPM_idle}, b_1 v_\mathrm{mph} + b_0) \quad (4)$$

$$\mathrm{qf}(v_\mathrm{mph}) = \max(0, c_1 v_\mathrm{mph} + c_0) \quad (5)$$

TABLE III
MEAN RPM AND FUEL MODEL PARAMETERS

Phase	b_1	b_0	c_1	c_0
J accel	55.987	1990.9	3.981e-05	6.538e-04
J brake	62.455	466.3	-1.831e-11	1.434e-04
C accel	23.742	1625.7	1.367e-05	3.285e-04
C brake	21.340	595.4	-2.682e-06	1.084e-04

b_1 is the RPM–speed slope (rpm/mph), and b_0 its intercept (rpm). c_1 is the fuel-rate–speed slope ((US gal/s)/mph), and c_0 its intercept (US gal/s). RPM_idle is the estimated idle engine speed (rpm), used as a lower limit. These are moving-phase fits, not measured idle fuel rates. Coefficients are rounded for display only.

Figs. 8–9 illustrate the RPM and fuel-rate models for Course 2. The RPM relationship is shown against speed during acceleration; the fuel-rate curve is evaluated along the predicted speed history.

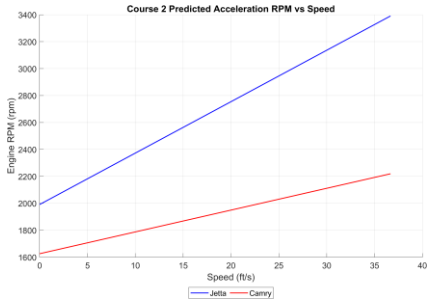


Figure 8. Course 2 Predicted Acceleration RPM vs. Speed. Predicted RPM-speed relationship during the acceleration phase of Course 2.

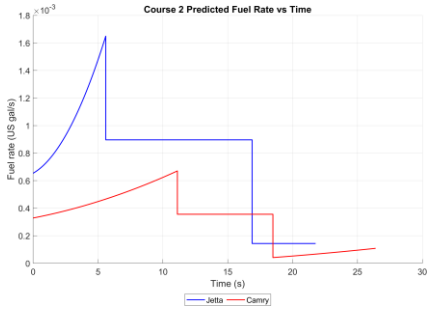


Figure 9. Course 2 Predicted Fuel Rate vs. Time. Predicted instantaneous fuel-use rate of both vehicles throughout Course 2.

For these low-speed runs, operations are mainly in first and second gear. A sudden RPM drop while vehicle speed continues to rise suggests an upshift. For example, Jetta Trial 1 shows a possible shift near 2950 rpm: RPM falls to 2587 while speed rises from 13.05 to 14.29 mph. This does not establish a fixed shift RPM or identify the gear. Since gear position was not logged, our acceleration and braking fits describe overall phase behavior rather than separate verified gear models.

Jetta moving fuel rate is v_mph/(3600 × MPG) where the logged MPG is positive. Camry fuel is converted from lb/min to lb/s, then divided by an estimated gasoline density of 6.17 lb/US gal [5]. Thus, both prediction sets report US gal/s, US gal, and MPG. If a trial fuel fit is negative at 0 or 25 mph, the code substitutes that phase's average rate. Predicted fuel rates are limited to nonnegative values. The Jetta RPM curve has a 928.21 idle rpm; the Camry prediction uses 685.42 idle rpm. During cruises, RPM and fuel rate are the midpoint of the acceleration and braking fits evaluated at the cruise speed. Substituting predicted speed into these relationships gives RPM

and fuel rate versus time. Cumulative fuel and instantaneous fuel economy follow from:

$$F(t) = \int_0^t qf(s)\,ds \quad (6)$$

$$MPG(t) = v(t)/(5280qf(t)),\ qf > 0 \quad (7)$$

Here t is time since launch (s), and s is the integration-time variable (s). qf(t) is the instantaneous fuel rate (US gal/s) evaluated at the predicted speed at time t; F(t) is accumulated fuel (US gal), with F(0) = 0. MPG(t) is instantaneous fuel economy (mi/US gal). Dividing v(t) in ft/s by 5280 gives mi/s; dividing by qf(t) gives mi/US gal. MPG is undefined when qf = 0.

Fuel is accumulated with cumtrapz() using the prediction timestamps.

IV. Course Predictions

Course 1 accelerates from rest to 15 mph, then cruises until x = 300 ft. Course 2 accelerates to 25 mph, cruises, then brakes to rest at x = 600 ft. MATLAB roots solves Eq. (2) for the earliest positive real acceleration time and, for Course 2, the braking time at v = 0. Let D be total course distance, Da acceleration distance, and Db braking distance (all in ft); vc is cruise speed (ft/s). The remaining cruise duration tc (s) is:

$$tc = (D - Da - Db)/vc \quad (8)$$

For Course 1, D = 300 ft and Db = 0; for Course 2, D = 600 ft and Db is obtained by integrating the braking model.

Course 3 covers 1320 ft with an overall average speed of 15 mph, so total time is 1320/22 = 60 s. It uses acceleration followed by cruise. The acceleration duration ta is obtained by solving:

$$x(ta) + v(ta)(60 - ta) = 1320 \quad (9)$$

Here ta is acceleration duration (s), x(ta) is the distance reached at the end of acceleration (ft), and v(ta) is the resulting cruise speed (ft/s). Thus the two terms are acceleration distance and the distance covered at constant speed during the remaining 60 − ta seconds.

This gives cruise speeds of 15.67 mph for Jetta and 16.21 mph for Camry: slightly above the required average to account for the slower launch. No braking phase is added to this course. Course 4 keeps accelerating for 1320 ft; its finish time is the earliest positive real root of:

$$(j/6)t^3 + (a_0/2)t^2 - 1320 = 0 \quad (10)$$

In Eq. (10), t is the finish time from launch (s), with $x_0 = 0$ and $v_0 = 0$. The mean acceleration-phase a_0 and j are used, and 1320 ft is the prescribed quarter-mile distance.

Selected Course 2 predictions are shown with the model equations in Figs. 5–9. The appendix includes the remaining predictions for all four courses, including motion, RPM, fuel, and phase-separated idle graphs. All prediction curves use the mean parameters in Tables II–III, with Jetta in blue and Camry in red.

V. Results and Conclusion

Tables IV–V summarize the model predictions. Courses 1 and 2 leave room for cruising after acceleration and any braking. Course 3 meets its specified average by construction.

TABLE IV
PREDICTED COURSE TIMING

Case	Accel (s)	Cruise (s)	Brake (s)	Total (s)
1 J	4.12	12.01	0.00	16.12
1 C	7.64	10.22	0.00	17.86
2 J	5.60	11.28	4.89	21.77
2 C	11.12	7.36	7.92	26.40
3 J	4.23	55.77	0.00	60.00
3 C	8.10	51.90	0.00	60.00
4 J	15.80	0.00	0.00	15.80
4 C	25.65	0.00	0.00	25.65

Case = course number and vehicle. J = Jetta; C = Camry.

TABLE V
PREDICTED FINAL CONDITIONS AND FUEL USE

Case	Final speed (mph)	Max accel (ft/s²)	Fuel used (US gal)
1 J	15.00	8.70	0.01203
1 C	15.00	3.80	0.00628
2 J	0.00	11.11	0.01661
2 C	0.00	4.63	0.00849
3 J	15.67	8.88	0.04343
3 C	16.21	3.91	0.01942
4 J	160.14	27.73	0.04616
4 C	88.08	8.11	0.02073

For Course 4, extending the same linear acceleration law gives 15.80 s and 160.14 mph for Jetta, and 25.65 s and 88.08 mph for Camry. This is an unrestricted continuation of a low-speed fit, not a claim of achievable quarter-mile performance. Cruise rates and the fuel-density conversion are estimates. Within those assumptions, the workflow consistently converts measured trials into phase equations and then into complete course predictions.

Fitting acceleration first keeps speed and distance mathematically connected through integration. Averaged coefficients provide one reusable model per car; separate moving phases retain distinct RPM and fuel behavior.

For raw-data reporting, we use an assumed rounding bound of plus or minus half the last reported increment. For example, if speed is rounded to 0.001 mph, 24.002 mph is reported as 24.002 ± 0.0005 mph. This describes rounding only, not verified sensor accuracy; extra digits from unit conversion do not establish a finer sensor resolution. The same rounding bound cannot be assigned directly to fitted or predicted quantities. Total prediction uncertainty is not quantified here.

References

[1] Mt. San Antonio College, Epic Quest 1d, Project 1 Overview v2, EpicQ1 grading rubric, and virtual-course instructions. Course materials.

[2]DataParser.mat, DataParser.m, DataModle.m, ModelPrediction.m, ModelPredictionPlotting.m, and the current trial and prediction PNG files. September 27, 2026.

[3] MathWorks, “Linear Kalman Filters.” Source

[4] MathWorks, MATLAB documentation for polyfit, polyval, polyint, roots, and cumtrapz(). Source | polyval | polyint | roots | cumtrapz()

[5] Atlantic Shores Offshore Wind, Appendix II-C, Air Emissions Calculation Methodology, PDF p. 6. Gasoline density assumption of 6.17 lb/gal. Hosted by BOEM. Source

[6] Oregon State University, “Lecture 1 | Graphical Analysis,” *BoxSand – Flip the Classroom*, Department of Physics, College of Science. Accessed Sept. 27, 2026

[7] DieselNet, “Central Business District (CBD) Driving Cycle,” *Emission Test Cycles*. Accessed Sept. 27, 2026.

Figure appendix: the following pages contain the remaining 14 trial figures and 36 prediction figures. Together with the selected figures in the main text, they cover motion, RPM, fuel, MPG, and phase-separated RPM versus speed. Full-resolution PNGs are retained in “New plots and reports.”

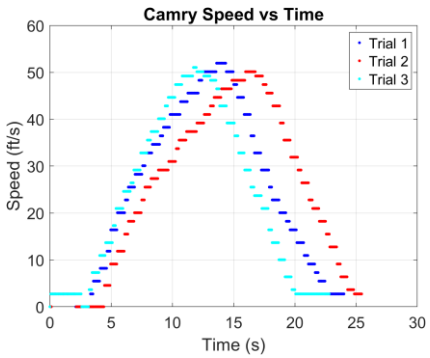


Figure 10. Camry Speed vs. Time. Recorded vehicle speed for the three Camry trials during acceleration and deceleration.

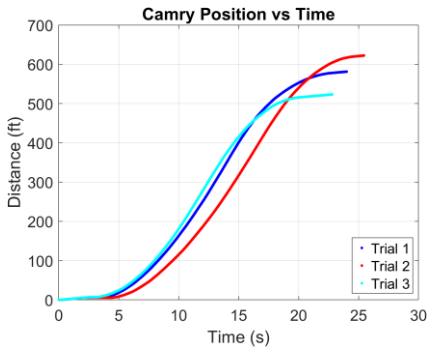


Figure 11. Camry Position vs. Time. Calculated position of the Camry over time for the three experimental trials.

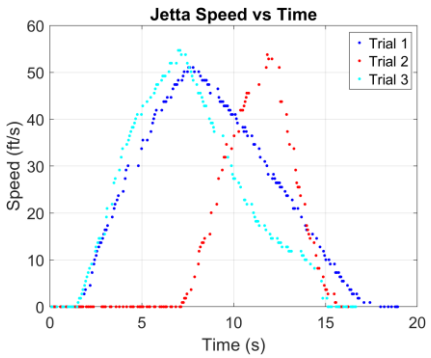


Figure 12. Jetta Speed vs. Time. Recorded vehicle speed for the three Jetta trials during acceleration and deceleration.

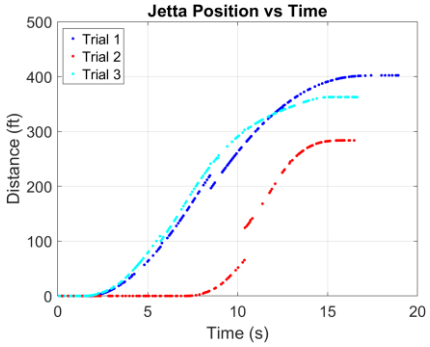


Figure 13. Jetta Position vs. Time. Calculated position of the Jetta over time for the three experimental trials.

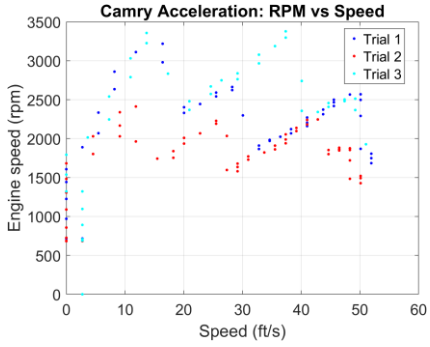


Figure 14. Camry Acceleration RPM vs. Speed. Relationship between engine RPM and vehicle speed while the Camry was accelerating.

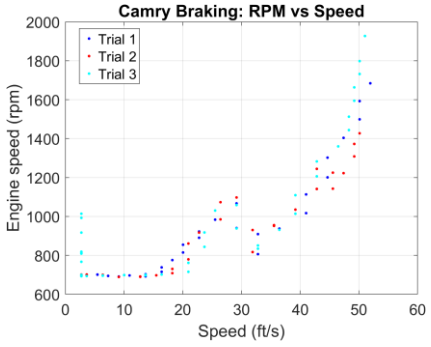


Figure 15. Camry Braking RPM vs. Speed. Relationship between engine RPM and vehicle speed while the Camry was braking.

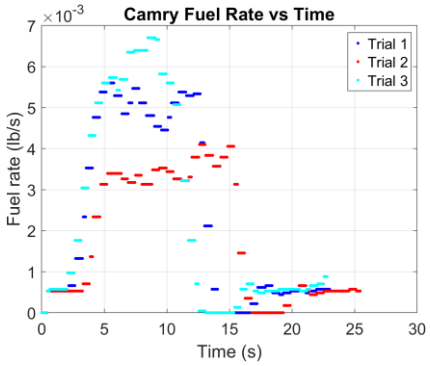


Figure 16. Camry Fuel Rate vs. Time. Fuel-use rate of theCamry throughout the three recorded trials.

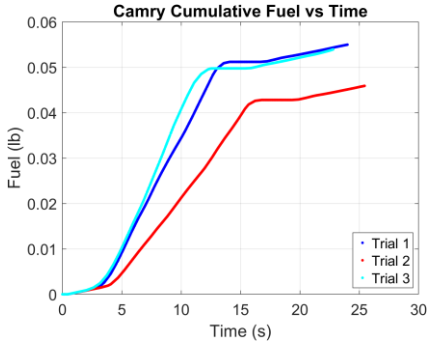


Figure 17. Camry Cumulative Fuel vs. Time. Total fuel used by the Camry as a function of time during each trial.

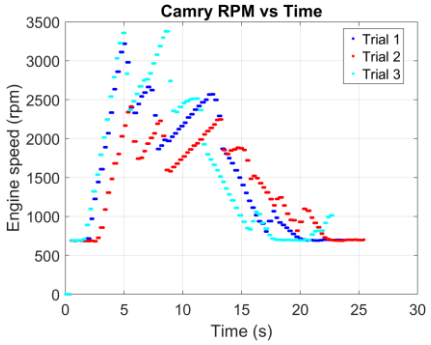


Figure 18. Camry RPM vs. Time. Engine RPM of the Camry over time during acceleration and deceleration.

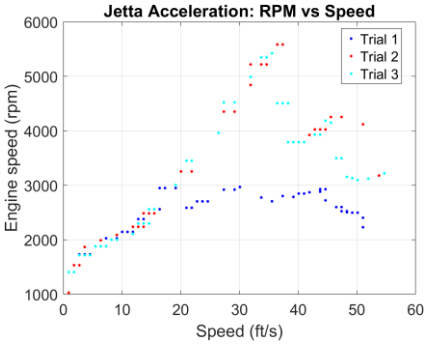


Figure 19. Jetta Acceleration RPM vs. Speed. Relationship between engine RPM and vehicle speed while the Jetta was accelerating.

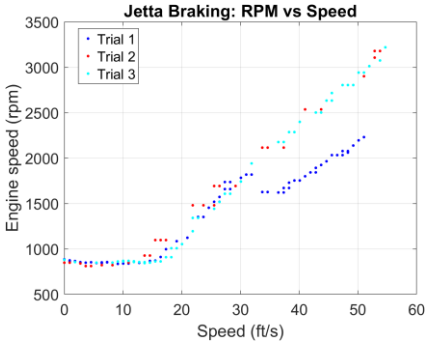


Figure 20. Jetta Braking RPM vs. Speed. Relationship between engine RPM and vehicle speed while the Jetta was braking.

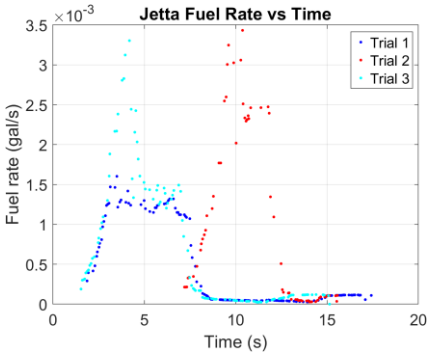


Figure 21. Jetta Fuel Rate vs. Time. Fuel-use rate of the Jetta throughout the three recorded trials.

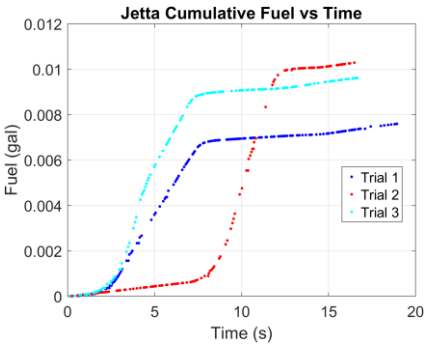


Figure 22. Jetta Cumulative Fuel vs. Time. Total fuel used by the Jetta as a function of time during each trial.

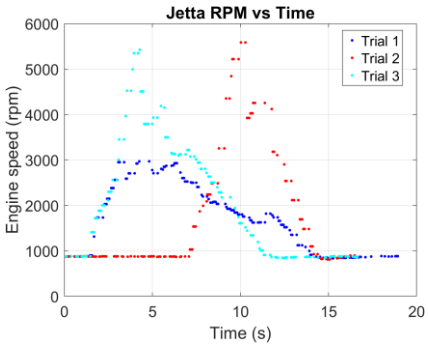


Figure 23. Jetta RPM vs. Time. Engine RPM of the Jetta over time during acceleration and deceleration.

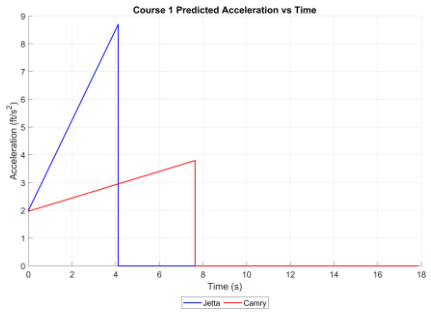


Figure 24. Course 1 Predicted Acceleration vs. Time. Predicted Jetta and Camry acceleration throughout Course 1.

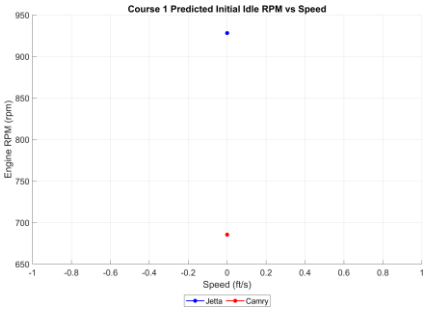


Figure 25. Course 1 Predicted Initial Idle RPM vs. Speed. Estimated engine RPM at zero vehicle speed for Jetta and Camry.

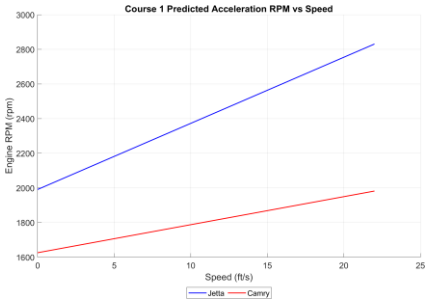


Figure 26. Course 1 Predicted Acceleration RPM vs. Speed. Predicted RPM-speed relationship during the acceleration phase of Course 1.

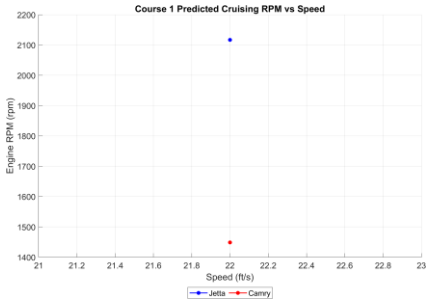


Figure 27. Course 1 Predicted Cruising RPM vs. Speed. Predicted engine RPM at the cruising speed used in Course 1.

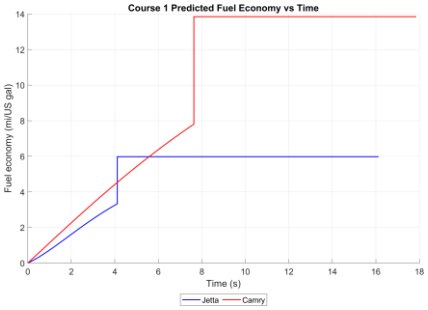


Figure 28. Course 1 Predicted Fuel Economy vs. Time. Predicted fuel economy of the Jetta and Camry throughout Course 1.

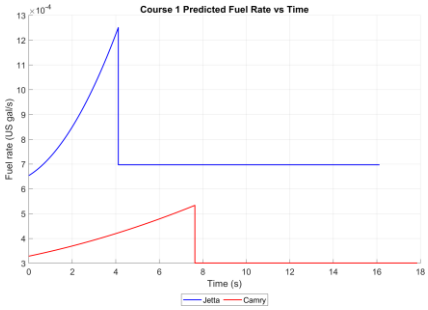


Figure 29. Course 1 Predicted Fuel Rate vs. Time. Predicted instantaneous fuel-use rate of both vehicles during Course 1.

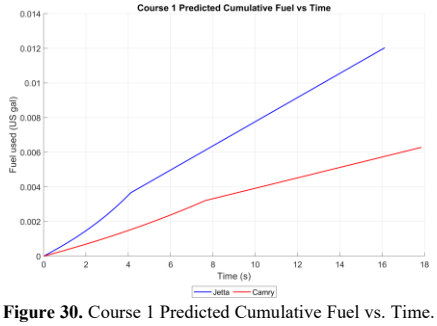


Figure 30. Course 1 Predicted Cumulative Fuel vs. Time. Predicted total fuel consumed by both vehicles during Course 1.

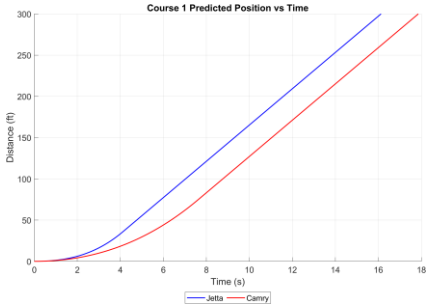


Figure 31. Course 1 Predicted Position vs. Time. Predicted distance traveled by the Jetta and Camry during Course 1.



Figure 32. Course 1 Predicted RPM vs. Time. Predicted engine RPM of both vehicles throughout Course 1.

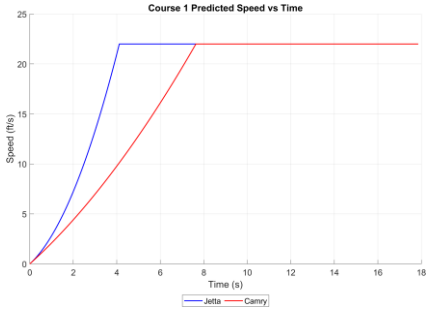


Figure 33. Course 1 Predicted Speed vs. Time. Predicted vehicle speed during the acceleration and cruising phases of Course 1.

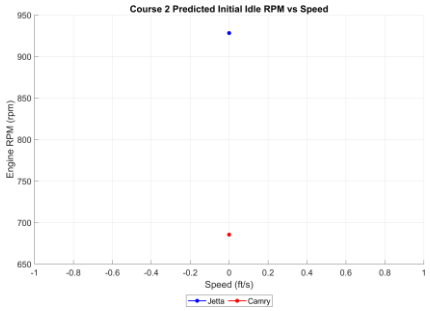


Figure 34. Course 2 Predicted Initial Idle RPM vs. Speed. Estimated engine RPM at zero vehicle speed for Jetta and Camry.

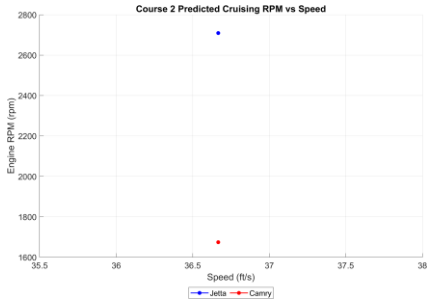


Figure 35. Course 2 Predicted Cruising RPM vs. Speed. Predicted engine RPM at the cruising speed used in Course 2.

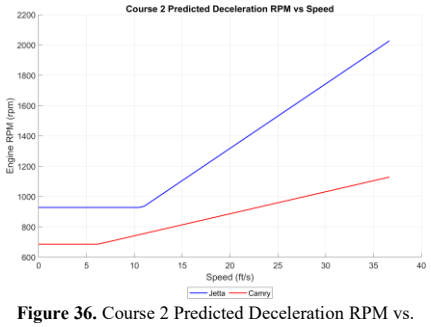


Figure 36. Course 2 Predicted Deceleration RPM vs. Speed. Predicted RPM-speed relationship during the braking phase of Course 2.

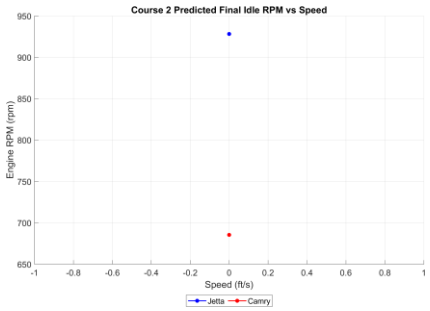


Figure 37. Course 2 Predicted Final Idle RPM vs. Speed. Estimated engine RPM at zero vehicle speed for Jetta and Camry.

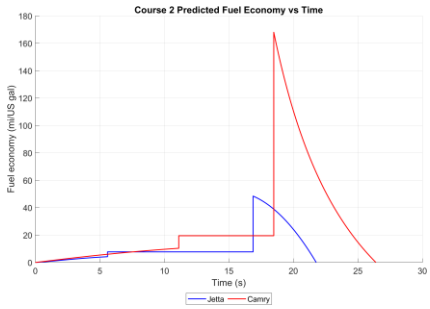


Figure 38. Course 2 Predicted Fuel Economy vs. Time. Predicted fuel economy of the Jetta and Camry throughout Course 2.

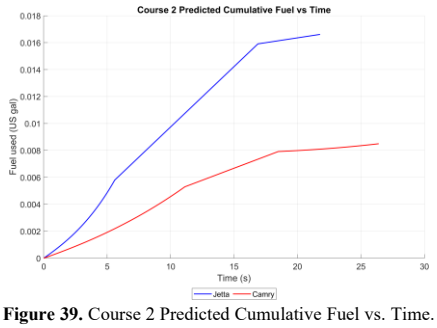


Figure 39. Course 2 Predicted Cumulative Fuel vs. Time. Predicted total fuel consumed by both vehicles during Course 2.

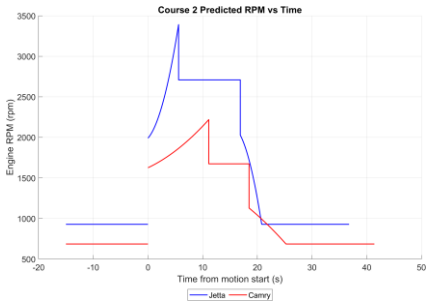


Figure 40. Course 2 Predicted RPM vs. Time. Predicted engine RPM of both vehicles throughout Course 2.

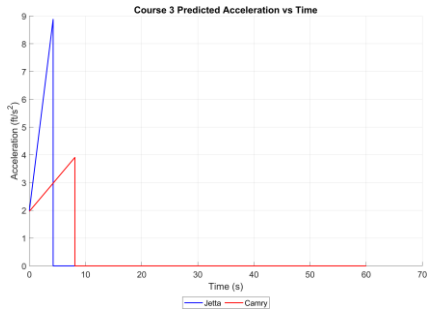


Figure 41. Course 3 Predicted Acceleration vs. Time. Predicted acceleration of the Jetta and Camry during Course 3.

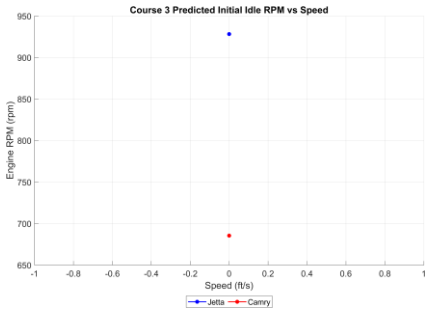


Figure 42. Course 3 Predicted Initial Idle RPM vs. Speed. Estimated engine RPM at zero vehicle speed for Jetta and Camry.

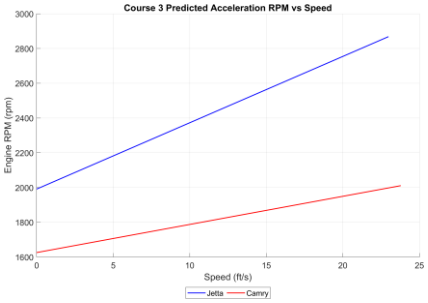


Figure 43. Course 3 Predicted Acceleration RPM vs. Speed. Predicted RPM-speed relationship during the acceleration phase of Course 3.

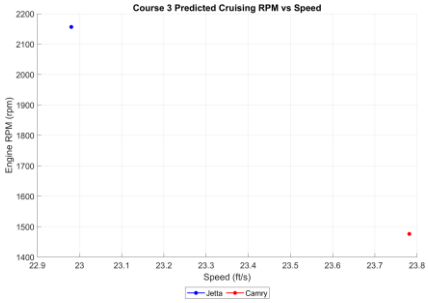


Figure 44. Course 3 Predicted Cruising RPM vs. Speed. Predicted engine RPM at the cruising speed used in Course 3.

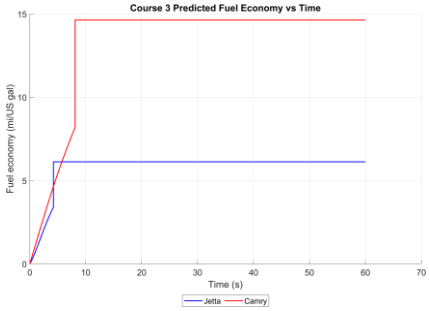


Figure 45. Course 3 Predicted Fuel Economy vs. Time. Predicted fuel economy of the Jetta and Camry throughout Course 3.

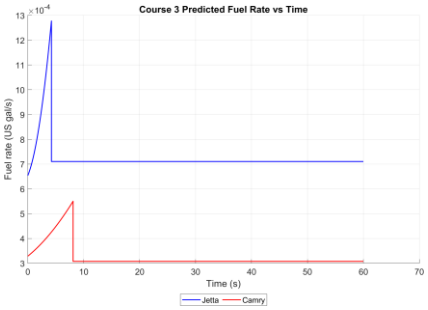


Figure 46. Course 3 Predicted Fuel Rate vs. Time. Predicted instantaneous fuel-use rate of both vehicles throughout Course 3.

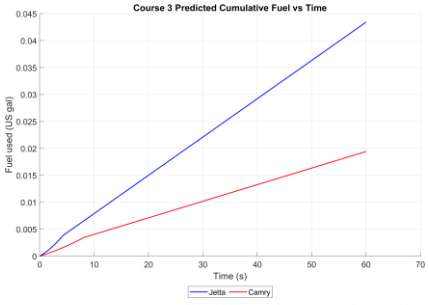


Figure 47. Course 3 Predicted Cumulative Fuel vs. Time. Predicted total fuel consumed by both vehicles during Course 3.

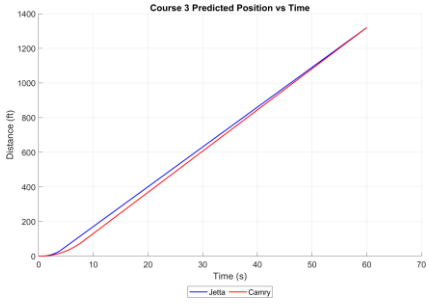


Figure 48. Course 3 Predicted Position vs. Time. Predicted distance traveled by the Jetta and Camry during Course 3.



Figure 49. Course 3 Predicted RPM vs. Time. Predicted engine RPM of both vehicles throughout Course 3.

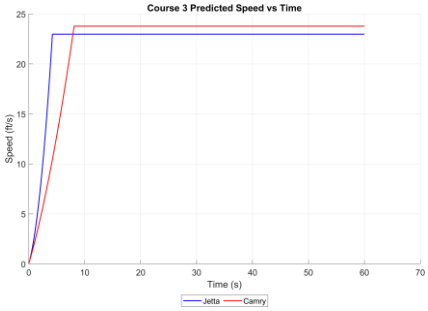


Figure 50. Course 3 Predicted Speed vs. Time. Predicted vehicle speed during the acceleration and cruising phases of Course 3.

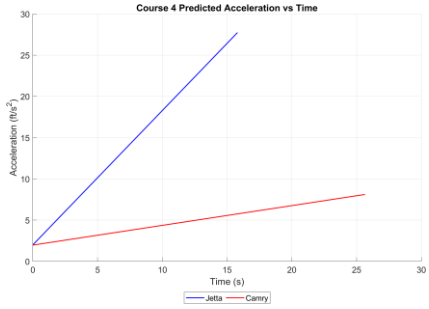


Figure 51. Course 4 Predicted Acceleration vs. Time. Predicted acceleration of the Jetta and Camry throughout Course 4.

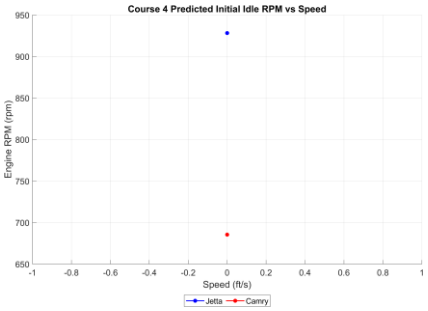


Figure 52. Course 4 Predicted Initial Idle RPM vs. Speed. Estimated engine RPM at zero vehicle speed for Jetta and Camry.

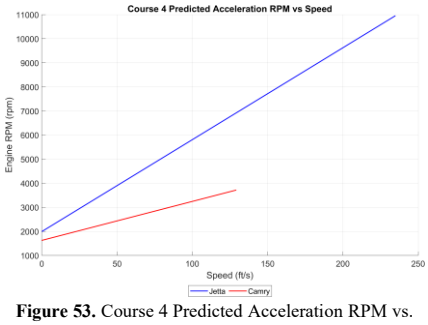


Figure 53. Course 4 Predicted Acceleration RPM vs. Speed. Predicted RPM-speed relationship while both vehicles accelerate in Course 4.

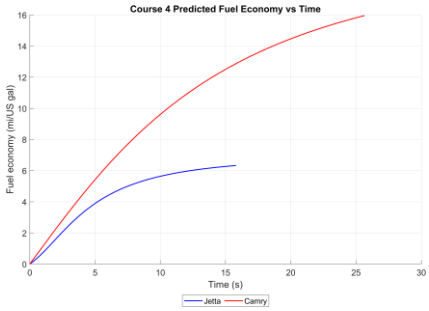


Figure 54. Course 4 Predicted Fuel Economy vs. Time. Predicted fuel economy of the Jetta and Camry throughout Course 4.

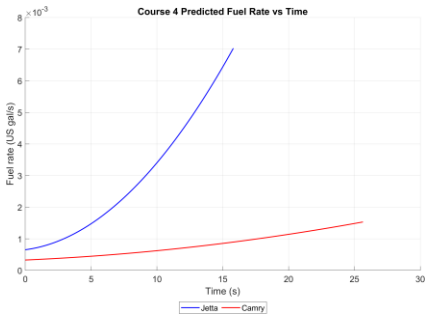


Figure 55. Course 4 Predicted Fuel Rate vs. Time. Predicted instantaneous fuel-use rate of both vehicles throughout Course 4.

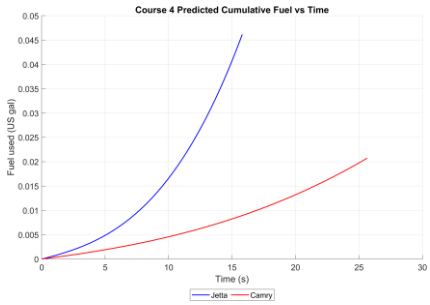


Figure 56. Course 4 Predicted Cumulative Fuel vs. Time. Predicted total fuel consumed by both vehicles during Course 4.

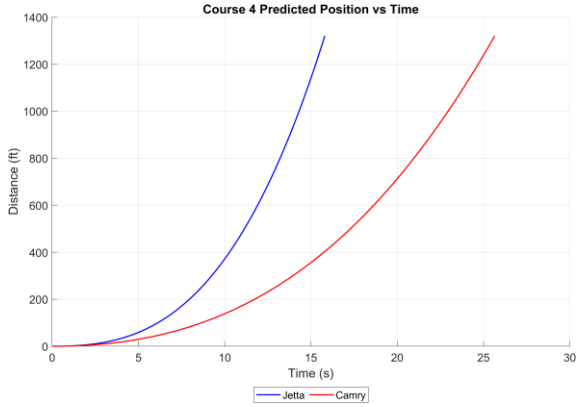


Figure 57. Course 4 Predicted Position vs. Time. Predicted distance traveled by the Jetta and Camry during Course 4.

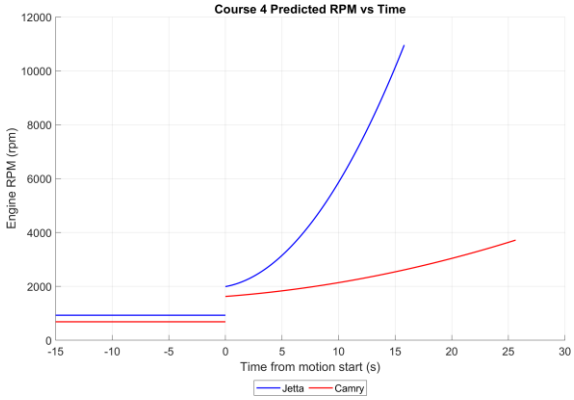


Figure 58. Course 4 Predicted RPM vs. Time. Predicted engine RPM of both vehicles throughout Course 4.

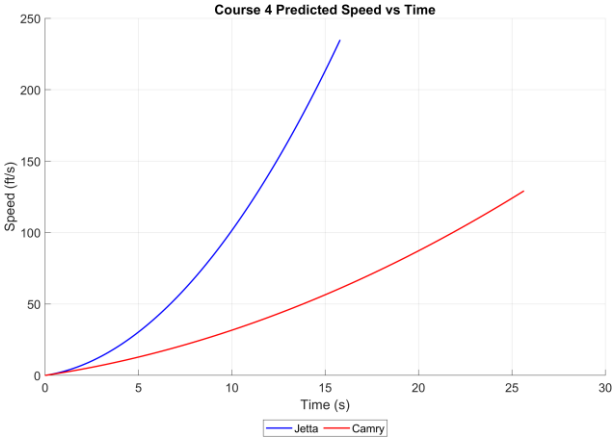


Figure 59. Course 4 Predicted Speed vs. Time. Predicted vehicle speed as both vehicles accelerate through Course 4.

Videos: https://drive.google.com/file/d/1BLSNiZdyAw1T6se-nYA1MQYnj5-Bsott/view?usp=sharing (Interior view)
https://drive.google.com/file/d/1lgTc8Egnt4QmQACJCArSgYihQx_k8Ipq/view?usp=sharing (exterior view)